Saketh Chokkaraju

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Education

Westfield High School — Herndon, VA
Expected Graduation: May 2029 — *Aug. 2025 – Present*

- Relevant Coursework: AP CSP, AP Human Geography, AP CSA, AP Calculus BC
- Clubs: TSA, Competitive Coding Club, Math Olympiad, Public Forum Debate

Experience

Lead Programmer — 2026 – Present
FTC Robotics Team — *Herndon, VA*

- Wrote autonomous and tele-op control code in Java; integrated sensor fusion and computer vision for navigation
- Collaborated with mechanical and electrical subteams to iterate on robot design and resolve hardware-software conflicts

Event Team Lead – Geospatial Technologies — 2026
Technology Student Association (TSA) — *Westfield High School*

- Led DMV Urban Heat Island analysis using Landsat 9, Census/CDC/NLCD data, QGIS, and ML models in Python
- Managed team coordination and presentation delivery – earned 1st Place at state competition

Deputy Chief of Staff, Research Team — 2026 – Present
Civil Air Patrol — *Herndon, VA*

- Lead cadets in aerospace and STEM research; coordinate logistics and mentor junior cadets on technical methodology
- Directing a High Altitude Balloon Challenge experiment analyzing piezoelectric energy output at altitude

Projects

MedStack | *Python, Groq Vision, Firecrawl* — 2026

- Built a medication safety app using Groq Vision and Firecrawl to scan pill bottles, extract drug info, and flag interactions with green/yellow/red severity ratings
- Incorporated clinical feedback from VCU's Division of Geriatric Medicine to align severity ratings with established drug interaction standards and medical guidelines

mireye-map | *Next.js 16, Mapbox GL JS, Mireye API* — 2026

- Built a full-stack interactive map (Next.js, TypeScript, Mapbox GL JS) that turns Mireye's JSON-only API into a tool for exploring cited FEMA/USGS/NOAA physical-world data across 6 presets.
- Delivered Mireye's visual product layer, letting users explore physical-world risk data and supporting the Mireye's parcel-level data roadmap.

Honors & Awards

1st Place – TSA Geospatial Technologies (Technosphere) | *Geospatial ML analysis of DMV Urban Heat Islands*
1st Place – TSA Technosphere (System Control) | *Built and programmed autonomous robotic systems*
Best Use of Firecrawl – TillyHacks | *Built drug compatibility web app using Firecrawl API*
Top 10 Finalist – DesignXR Hackathon | *Developed a VR game; top 10 finalist internationally*
2nd Place – Fairfax County Regional Science Fair (Plant Sciences) | *Experimental plant growth research*
3rd Place – AI Playlab Challenge | *Created AdmitPath, an AI extracurricular recommendation agent*
1st Place School Winner – CEMC Pascal Math Contest | *Certification of Distinction, Top 20% of contestants*

Technical Skills

Programming/Web: Python, Java, HTML/CSS, Next.js
Engineering: Autodesk Fusion 360, Robotics System Control
Software: Google Workspace, Google Sheets, Git
Geospatial: QGIS, GeoPandas, Google Earth, Landsat Imagery, ArcGIS Online
AI/ML: LLMs, AI Pipelines, AI agents, RAG, MCPs, and Multi-Agent Systems